

EXPERIMENT 6

RAIL FENCE CIPHER

AIM

To implement the Rail Fence transposition technique using C.

ALGORITHM

1. Read the plaintext.
2. Arrange the characters in zig-zag order.
3. Read the rows sequentially to obtain the cipher text.
4. Display the encrypted text.

CODE

```
#include <stdio.h>
#include <string.h>

int main() {
    char text[100];
    int rails;

    printf("Enter Plain Text: ");
    scanf("%s", text);

    printf("Enter Number of Rails: ");
    scanf("%d", &rails);

    int len = strlen(text);
    char rail[10][100];
    int i,j;

    for(i=0;i<rails;i++)
        for(j=0;j<len;j++)
            rail[i][j]='\n';

    int row=0,dir=1;

    for(i=0;i<len;i++){
        rail[row][i]=text[i];

        if(row==0)
            dir=1;
        else if(row==rails-1)
            dir=-1;
    }
```

```

        row+=dir;
    }

    printf("Encrypted Text: ");

    for(i=0;i<rails;i++)
        for(j=0;j<len;j++)
            if(rail[i][j]!='\n')
                printf("%c",rail[i][j]);

    return 0;
}

```

OUTPUT

Output

Clear

```

Enter Plain Text: meetmeatthepartytonight
Enter Number of Rails: 3
Encrypted Text: mmtatgetethpryoiehaetnt

=== Code Execution Successful ===

```

RESULT

Thus the Rail Fence transposition technique was implemented successfully using C.